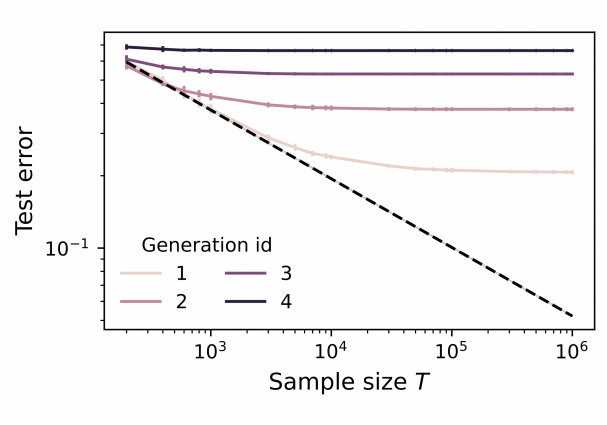

Figure 15. **Empirical Hutter++ Model.** Same setting as in Figure 14 with top $p^{inf} = 0.9$ synthesizing. No temperature scaling is used. $\beta = 3/2$. Top-p^{inf} selection significantly deteriorate the model collapse.

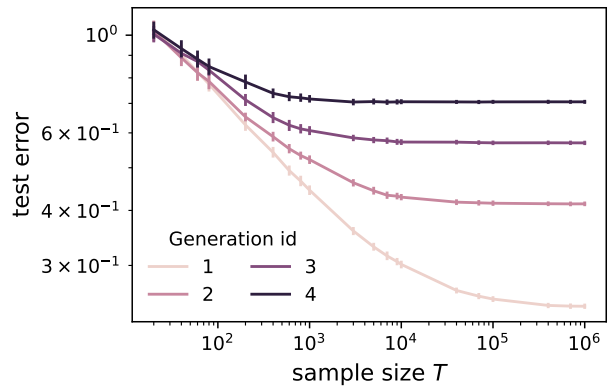

Figure 16. **Empirical Hutter++ Model.** Same setting as in Figure 14 with temperature τ =0.9 synthesizing. No top-p^{inf} selection is used. $\beta = 3/2$. Compared with Figure 14, temperature also create strong model collapse across multiple generation.

Also $u(1) = T$ and $u(k) = Tk^{-\beta}$. We deduce that

$$\begin{aligned}
\sum_{i=1}^{k} p_i e^{-Tp_i} &\asymp \int_1^k x^{-\beta} e^{-Tx^{-\beta}} \mathrm{d}x = \int_{Tk^{-\beta}}^{T} (u/T) e^{-u} (T^{1/\beta} u^{-1-1/\beta}/\beta) \mathrm{d}u \\
&\asymp T^{-(1-1/\beta)} \int_{Tk^{-\beta}}^{T} u^{-1/\beta} e^{-u} \mathrm{d}u \\
&\asymp T^{-(1-1/\beta)} \left(\Gamma(1-1/\beta, Tk^{-\beta}) - \Gamma(1-1/\beta, T) \right) \\
&= T^{-c} \left(\Gamma(c, Tk^{-\beta}) - \Gamma(c, T) \right),
\end{aligned}$$

and we are done for the first part.

For the second part, note that $\Gamma(c, T) = o(1)$ for large T so that

$$(\Gamma(c, Tk^{-\beta}) - \Gamma(c, T))|_{k=\infty} = \Gamma(c, 0) - \Gamma(c, T) = \Theta(1) - o(1) = \Theta(1),$$

from which the result follows. □

We now consider two separate cases for the relative scaling of k and T.

– Case 1: $T \gtrsim k^{\beta}$**.** Here, we have thanks to Lemma C.1

$$E_{test} \asymp k^{-(\beta-1)} + O(T^{-c}) \asymp k^{-(\beta-1)}, \tag{23}$$

since $k^{-(\beta-1)} \gtrsim T^{-(\beta-1)/\beta} = T^{-c}$.

– Case 2: $1 \ll T \lesssim k^{\beta}$**.** Here, thanks to Lemma C.1 we have $\Gamma(c, T) = o(1)$ and $\Gamma(c, Tk^{-\beta}) = \Theta(1)$. We deduce that

$$E_{test} \asymp k^{-(\beta-1)} + T^{-c} \left(\Gamma(c, Tk^{-\beta}) - \Gamma(c, T) \right) \asymp k^{-(\beta-1)} + T^{-c} \asymp T^{-c}, \tag{24}$$

since $k^{-(\beta-1)} \lesssim T^{-(\beta-1)/\beta} = T^{-c}$. Putting things together then gives the claimed result. □

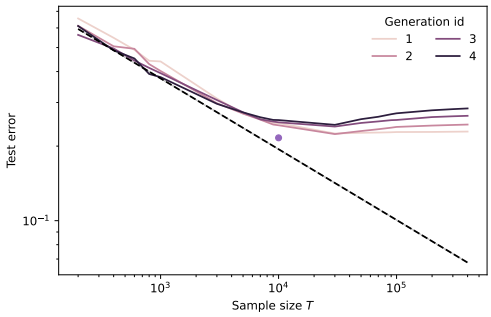

Figure 17. **Empirical Hutter++ Model with Mixing.** The initial "clean" dataset comprises $T_0 = 10,000$ samples. For future generations, the largest model is used to synthesize data. For $T \leq 20,000$, training data is an equal mix of clean and generated data, for $T > 20,000$ all clean data is used; the remaining training data is synthetic (so the ratio of clean data diminishes). Top-p^{inf} =0.9, no temperature scaling, $\beta = 3/2$.

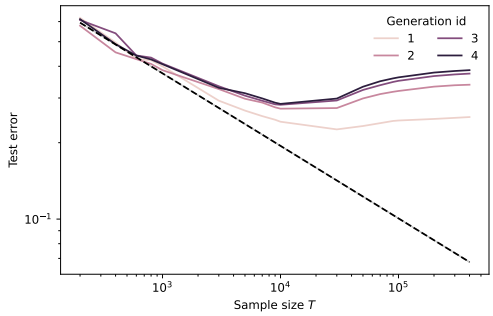

Figure 18. **Empirical Hutter++ Model with Mixing.** Same setting as in Figure 17 with top-$p^{inf} = 0.9$, no temperature scaling, and $\beta = 3/2$.

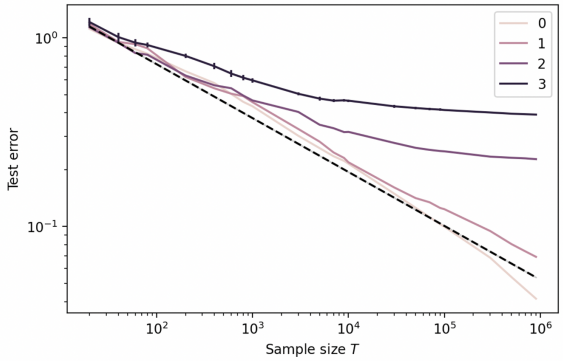

Figure 19. **S-shape "Smoothed Grokking".** Bigram data with Hutter++ model, mixing clean data with AI generated data with ratio 50 to 50. The grokking line is smoothed in the probabilistic setting. Line 1, 2, 3 are generated by using 10,000, 1,000, and 100 data to train the generating model. Compared to Figure 17, we do not limit the number of accessible real data now. $\beta = 3/2$.

C.2. Proof of Corollary 2.3

Indeed, let $p_i \propto i^{-\beta}$ and $(p_{AI})_i = q_i \propto i^{-\beta'}$. Then,

$$E_{test} \asymp \sum_i p_i(1-q_i)^T \asymp \sum_i p_i e^{-q_i T} \asymp \int_1^\infty x^{-\beta} e^{-x^{-\beta'}T} \mathrm{d}x. \tag{25}$$

Setting $u = x^{-\beta'}T$ gives $x = T^{1/\beta'}u^{-1/\beta'}$, and so $\mathrm{d}x = -(T^{1/\beta'}/\beta')u^{-(1+1/\beta')}\mathrm{d}u$. We deduce that

$$\begin{aligned} E_{test} &\asymp T^{-(\beta-1)/\beta'} \int_1^T u^{\beta/\beta'} u^{-(1+1/\beta')} e^{-u} \mathrm{d}u = T^{-(\beta-1)/\beta'} \int_1^T u^{(\beta-1)/\beta'-1} e^{-u} \mathrm{d}u \\ &\asymp T^{-c}\Gamma(c,T) = T^{-c}(1+o(1)), \text{ with } c := (\beta-1)/\beta'. \end{aligned}$$

That is, $E_{test} \asymp T^{-c}$ as claimed. □

C.3. Proof of Theorem 3.2 and Corollary 3.3

Suppose that of T samples available for training our model, πT are samples from the true distribution $p = Zipf(\beta)$ and $(1-\pi)T$ are from AI data distribution p' which is a version of p with its tail chopped off at rank k, i.e such that

$p'_i \propto p_i 1[i \leq k]$. Thus the dataset is drawn from the distribution given by $q_i = \pi p_i + (1-\pi)p'_i$. Test error of a Hutter LLM then writes

$$\begin{aligned} E_{test} &= \sum_{i\geq 1} p_i(1-q_i)^T = \sum_{1\leq i\leq k} p_i(1-p_i)^T + \sum_{i\geq k+1} p_i(1-\pi p_i)^T \\ &\asymp \sum_{1\leq i\leq k} p_i e^{-p_i T} + \sum_{i\geq k+1} p_i e^{-\pi p_i T}. \end{aligned} \tag{26}$$

Now, thanks to Lemma C.1, it is clear that for any integers $1 \leq r < R \leq \infty$ and large z, one has

$$\sum_{r\leq i\leq R} p_i e^{-p_i z} \asymp z^{-c}\left(\Gamma(c, zR^{-\beta}) - \Gamma(c, zr^{-\beta})\right), \tag{27}$$

where $c = 1 - 1/\beta \in (0,1)$ and Γ is the (upper) incomplete gamma function. Applying (27) with $(r,k,z) = (1,k,T)$ gives

$$T^c \sum_{1\leq i\leq k} p_i e^{-p_i T} \asymp \Gamma(c, Tk^{-\beta}) - \Gamma(c,T) = \begin{cases} \Theta(1) - o(1) = \Theta(1), & \text{if } 1 \ll T \lesssim k^\beta, \\ o(1) - o(1) = o(1), & \text{if } T \gtrsim k^\beta \gg 1. \end{cases} \tag{28}$$

On the other hand, applying (27) with $(r,k,z) = (k+1, \infty, \pi T)$ and assuming $\pi = \Theta(1)$ gives

$$\sum_{i\geq k+1} p_i e^{-\pi p_i T} \asymp (\pi T)^{-c}\gamma(c, \pi T(k+1)^{-\beta}) \asymp \begin{cases} (\pi T)^{-c}, & \text{if } \pi T \gtrsim k^\beta \gg 1, \\ (k+1)^{-\beta c} \asymp k^{-\beta c}, & \text{if } k^\beta \gg \pi T. \end{cases} \tag{29}$$

Putting things together gives the result. □

Recall that Bertrand et al. (2023) also formally study such mixtures for iterative retraining. In their setting, they show the existence of fixed points in the mixture proportion that delineates the region of model collapse. These results are complimentary and not contradictory to ours: they combine mixing, large number of iteration, and data-decay, thus studying a combination of effects (under different theoretical conditions, not focusing on scaling laws) that our preceding theorems address separately.

C.4. Grokking for Tail Narrowing

Theorem C.2 (Grokking with Tail Narrowing)**.** *Consider a sample of size T of which a proportion π comes from the true distribution $p = Zip(\beta)$ and the remainder comes from a version $p' = Zip(\beta')$. We have the following scaling law for the Hutter LLM,*

$$E_{test} \asymp (\pi T)^{-c} + ((1-\pi)T^{-c'}), \tag{30}$$

where $c := (\beta-1)/\beta$ and $c' := (\beta-1)/\beta'$.

Define $\overline{T} := (\pi/(1-\pi))^{-a}$, where $a := s/(1-s)$, and $s := \beta/\beta'$. Then,

*(A) **Early-Stage Dynamics.** For $T \lesssim \overline{T}$, it holds that $E_{test} \asymp ((1-\pi)T)^{-c'}$. Thus, if $\beta' > \beta$, the money spent on acquiring some clean data is not amortized!*

*(B) **Later-Stage Dynamics.** As soon as $T \gtrsim \overline{T}$, it holds that $E_{test} \asymp (\pi T)^{-c}$. Similarly, we recover the unpolluted sample-size law scaling T^{-c}. For fixed T and tunable π, this error rate scales like π^{-c}.*

Proof. Let q be the mixture of p and p'. We prove the result for $\beta' \geq \beta$; the case $\beta' \leq \beta$ is analogous. So, one may write

$$E_{test} = \sum_{i\geq 1} p_i(1-q_i)^T \asymp \sum_{i\geq 1} p_i e^{-\pi i^{-\beta} + (1-\pi)i^{-\beta'}} \asymp \sum_{1\leq i\leq \overline{T}^{1/\beta}} p_i e^{-\pi i^{-\beta}} + \sum_{i\geq \overline{T}^{1/\beta}} p_i e^{-(1-\pi)i^{-\beta'}}, \tag{31}$$

where we have used the fact that $(1-\pi)i^{-\beta'} \geq \pi i^{-\beta}$ iff $i \leq (\pi/(1-\pi))^{-1/(\beta'-\beta)} = \overline{T}^{1/\beta}$. The result then follows from (27). □

Remark C.3. Let us conclude by saying that clean data always helps, since E_{test} is decreasing function of π. Indeed, from (26), the derivative w.r.t π is $E'_{test}(\pi) = -T\sum_{i\geq k+1} p_i^2(1-\pi p_i)^{T-1} \leq 0$.